1.9 Random Vectors

Random Vectors

Consider two random variables X_1 and X_2 both defined on the same probability space $\{\Omega, \mathcal{F}, \mathbb{P}\}$ and let $\mathcal{B}(\mathbb{R}^2)$ be the Borel algebra of $\mathbb{R}^2$. We define the random vector $X : \Omega \to \mathbb{R}^2$

$$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

such that

$$\textit{Pre-image}(B) = \boldsymbol{X}^{-1}(B) = \{\omega \in \Omega : \boldsymbol{X}(\omega) = x \in B\} = A \in \mathcal{F}$$

for any $B \in \mathcal{B}(\mathbb{R}^2)$.

Joint distributions

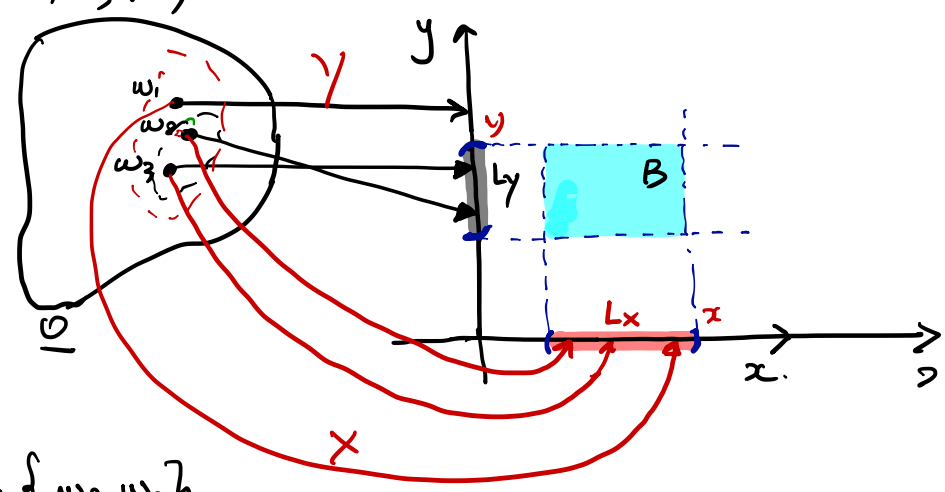
Joint Induced probability

$$B = (L_x, L_y) \in \mathcal{B}(\mathbb{R}^2)$$

$$P_{XY}(x \in L_x \text{ and } y \in L_y) = \mathbb{P}\left(X^{-1}(L_x) \cap Y^{-1}(L_y)\right)$$

$$\overset{\shortparallel}{P((x,y) \in (L_x, L_y))}$$

$$(\underline{\Omega}, \mathcal{F}, \mathbb{P}) \qquad (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2), P_{XY})$$



$$X^{-1}(L_x) = \{\omega_1, \omega_2, \omega_3\}$$

$$X^{-1}(L_y) = \{\omega_2, \omega_3\}$$

$$X^{-1}(L_x) \cap Y^{-1}(L_y) = \{\omega_2, \omega_3\}$$

Joint pdf

Let $B = [a_1, b_1] \times [a_2, b_2)$

- $f_{\underline{X}}(\underline{x}) \geq 0$
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\underline{X}}(x_1, x_2)\, dx_1\, dx_2 = 1$

$$P(\underline{x} \in B) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f_{\underline{X}}(x_1, x_2)\, dx_1\, dx_2$$

Joint CDF

$$F_{\underline{X}}(\underline{x}) = \mathbb{P}(X_1 \leq x_1 \text{ and } X_2 \leq x_2)$$

$$= \int_{-\infty}^{x_1} \int_{-\infty}^{x_1} f_{X_1, X_2}(x_1, x_2)\, dx_1\, dx_2$$

and $$f_{X_1, X_2}(x_1, x_2) = \frac{\partial F_{X_1, X_2}(x_1, x_2)}{\partial x_1 \partial x_2}$$

Marginal distributions

Marginal PDF

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1,x_2)\, dx_2 \quad \text{or}$$

$$f_{X_2}(x_2) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1,x_2)\, dx_2$$

Expectation

$$E(X_1) = \int_{-\infty}^{\infty} f_{X_1}(x_1)\, dx_1 = \mu_1$$

$$E(X_2) = \int_{-\infty}^{\infty} f_{X_2}(x_2)\, dx_2 = \mu_2$$

$$\underline{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} = E(\underline{X}).$$

Autocorrelation

$$E(X_1 X_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 x_2 f_{X_1X_2}(x_1,x_2)\, dx_1\, dx_2 = R_{X_1X_2}$$

Covariance

$$\mathrm{Cov}(X_1,X_2) = E\left((X_1 - E(X_1))(X_2 - E(X_2)) \right)$$

$$= E(X_1X_2) - E(X_1)E(X_2) = R_{X_1X_2} - \mu_1\mu_2.$$

Covariance is a measure of how changes in X are associated with changes in Y

It can be positive or negative.

- $Var(X_1+X_2) = Var(X_1) + Var(X_2) + 2\,Cov(X_1,X_2)$
- If X_1 and X_2 are independent, $Cov(X_1,X_2)=0$

  and

  $Var(X_1,X_2) = Var(X_1) + Var(X_2)$

Correlation. (normilized Covariance)

$$\rho_{x,y} = \frac{Cov(x,y)}{\sqrt{Var(x)\,Var(y)}} = \frac{Cov(x,y)}{\sigma_x\,\sigma_y}, \quad -1 \le \rho_{x,y} \le 1.$$

Covariance Matrix

$$C = \begin{pmatrix} C(X_1,X_1) & C(X_1,X_2) \\ C(X_2,X_1) & C(X_2,X_2) \end{pmatrix}$$

$$C(X_i,X_i) = Var(X_i), \quad i=1,2.$$

n-dimensional Random Vectors

Consider n random variables $X_1, X_2, \dots X_n$ all defined on the same probability space $\{\Omega, \mathcal{F}, \mathbb{P}\}$ and let $\mathcal{B}(\mathbb{R}^n)$ be the Borel algebra of $\mathbb{R}^n$. We define the random vector $X : \Omega \to \mathbb{R}^n$

$$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}$$

such that

$$\textit{Pre-image}(B) = \boldsymbol{X}^{-1}(B) = \{\omega \in \Omega : \boldsymbol{X}(\omega) = x \in B\} = A \in \mathcal{F}$$

for any $B \in \mathcal{B}(\mathbb{R}^n)$.

Joint cdf

$$F_{X_1,X_2,\dots,X_n}(x_1, x_2, \dots, x_n) = F_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(X_1 \le x_1, X_2 \le x_2, \dots, X_n \le x_n)$$

Joint pdf

If there exist a pdf function, i.e.,

$$f_{\mathbf{X}}(\mathbf{x}) = f_{X_1,X_2,\dots,X_n}(x_1, x_2, \dots, x_n)$$

then

$$F_{\mathbf{X}}(\mathbf{x}) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \cdots \int_{-\infty}^{x_n} f_{X_1,X_2,\dots,X_n}(s_1, s_2, \dots, s_n) ds_1 ds_2 \cdots ds_n$$

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